Averages hide the binding constraint

A current Bangladesh health experiment, a reform simulation for Uzbekistan, and a disaster-growth classic · two peer-reviewed articles and one World Bank working paper. The following article pages are edited reading excerpts, not complete replacements for the originals.

LITERATURE PULSE

These papers share a warning against policy conclusions drawn from averages. In rural Bangladesh, information changes what households know but not whom they visit; a first successful public-care experience changes behavior and spending. In Uzbekistan, complementary reforms lift simulated growth and lower prices, yet gains differ by region, skill, sector, and location while gender wage gaps persist. The disaster classic shows the same problem at macro scale: effects flip with disaster type, severity, sector, and national income. **Literature trend:** the useful question is increasingly not whether a policy or shock works on average, but which margin binds, who crosses it, and who remains exposed. Better policy pairs a credible behavioral or structural model with explicit distributional and heterogeneity checks.

01 CURRENT / GLOBAL HEALTH, TRUST, AND PROVIDER CHOICE · STARTS ON PAGE 2 · Health Policy and Planning · September 2026 · peer reviewed · CC BY 4.0

Evidence on Information, Access Frictions, and Experience in Public Healthcare Use in Rural Bangladesh: A Cluster-Randomized Field Experiment

Why do rural households keep using informal drug sellers when public care is cheap or free? A cluster-randomized trial with 2,320 Bangladeshi households separates information, access support, and a facilitated first public visit. Information raises awareness by 33 percentage points but does not change provider choice; the experience arm raises completed public visits by 18 points. Instrumental-variable estimates imply that induced public use reduces monthly health spending by about 601 taka (32%), mainly through medicines. The first stage is strong, but retrospective registration, self-reported use, a two-month follow-up, and possible overlap between trust and friction mechanisms limit interpretation. The result is a sharp test of whether the binding constraint is ignorance or uncertainty about an unfamiliar service.

02 CURRENT / REFORM SEQUENCING AND DISTRIBUTION · STARTS ON PAGE 8 · World Bank Policy Research Working Paper 11482 · October 2026 · open access

Distributional Effects of Market-Oriented Reforms in Uzbekistan: A Macro-Micro Simulation Approach

Who gains from Uzbekistan's trade, competition, preference, and investment reforms? The authors link the World Bank's MANAGE computable general-equilibrium model to a reweighted 2024 household survey, then trace employment, wage, price, poverty, and inequality effects through 2040. Coordinated reforms raise simulated GDP growth above the 5.8% baseline, and an FDI package adds further gains, while consumer prices fall and poverty declines unevenly across regions, skills, sectors, and locations. The framework is transparent and supported by a reproducibility package, but its conclusions are model- and calibration-based, not causal estimates; closure rules, elasticities, and assumed reform implementation matter. The value is distributional: an average growth dividend can coexist with persistent gender gaps and transitional losses.

03 CLASSIC / DISASTERS, SECTORAL GROWTH, AND HETEROGENEITY · STARTS ON PAGE 23 · World Development 40(7) · 2012 · peer reviewed classic · open World Bank edition

Natural Disasters and Growth: Going Beyond the Averages

Do disasters depress growth, or can reconstruction temporarily accelerate it? Using a dynamic panel of 94 countries from 1961–2005, the authors estimate system-GMM models separately by disaster type, economic sector, intensity, and income group. Average aggregate effects conceal large differences: droughts hurt agriculture, some moderate floods are followed by faster growth, and severe shocks erase or reverse favorable responses. Effects are especially heterogeneous in developing economies. The design is influential because it moved the literature beyond a single disaster dummy, though cross-country GMM, disaster measurement, and the exclusion restrictions are less persuasive than modern event-study or geospatial designs. The enduring lesson is to ask which sector, which shock, and how severe before interpreting an average effect.

Editorial note: references and appendices were omitted where possible; no included article contributes more than 15 text pages. Full citations and source links remain in the catalog. AI-written synopses were checked against the included source pages.

Evidence on information, access frictions, and experience in public healthcare use in rural Bangladesh: a cluster-randomized field experiment

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Abstract

In rural Bangladesh, many households continue to rely on private and informal providers even when public healthcare options are available at low or zero monetary cost. This paper studies whether the barrier is mainly misinformation, access frictions, or trust and uncertainty about public care. A cluster-randomized field experiment was conducted with 2,320 households across 580 neighborhoods and three sequential treatment arms. Information provision sharply increases awareness of public care options but has limited effects on actual utilization. In contrast, encouraging a one-time experience with public care produces significant increases in public uptake, substitution away from informal providers, and meaningful reductions in out-of-pocket spending, especially on medicines. Households induced to use public care reduced total health spending by approximately 601 Bangladeshi taka (USD 5.46 at 2023 exchange rates; 32% of the control-group mean) relative to nonusers (95% CI 373–829 taka; $P < 0.001$), with most of the reduction driven by lower medicine expenditure. The findings suggest that improving beliefs alone is insufficient and that lowering uncertainty through a first successful experience is a key lever for shifting healthcare demand toward public options.

Key messages

Information campaigns substantially raised awareness of free public healthcare services but produced no change in actual provider choice. A single facilitated experience with public care increased uptake by 18 percentage points and reduced total household health spending by 32%, driven by lower medicine expenditure and reduced exposure to overprescribing informal providers. The findings are consistent with healthcare functioning as both a credence good and an experience good: information alone cannot overcome habitual provider loyalty when patients have no direct basis for comparing public and informal care quality.

Introduction

In many low-income settings, households routinely seek medical advice from informal drug sellers even when certified health professionals are formally available. This pattern remains especially common in rural Bangladesh. Informal providers frequently recommend excessive or inappropriate treatments, particularly medicines, contributing to high out-of-pocket costs and antimicrobial resistance. Despite an expansion of community clinics, government facilities, and mobile health services, in-person public healthcare utilization remains strikingly low in rural areas.

The study tests three sequential demand-side interventions: information provision, practical access support, and encouragement to obtain a one-time assisted first use. The design asks whether households do not know about public care, face practical frictions in reaching it, or remain uncertain about an unfamiliar provider despite knowing it exists.

The literature identifies three distinct frictions. First, households may lack accurate knowledge of public facility locations, hours, fees, or provider qualifications. Second, travel time, queue uncertainty, and the cognitive burden of navigating bureaucratic systems may deter use. Third, because healthcare has credence-good characteristics, households rely on social networks and prior experience to form beliefs about provider quality. These frictions interact: uncertainty about public care may persist even after households receive factually accurate information.

The results show that awareness and behavior are decoupled. Information raised awareness by 33 percentage points but did not change provider choice. In contrast, a single experience of obtaining formal medical advice increased completed public visits by 18 percentage points and reduced spending among households induced to use public care. The paper argues that direct experience can update beliefs, reduce uncertainty, and break habitual provider choice.

Study setting and sample

Approximately 58% of rural households rely on local informal providers as their primary source of medical care, and medicine purchases account for about 60% of rural health expenditures. Patients cannot verify treatment appropriateness even after consumption, while they can readily observe convenience, familiarity, and interpersonal responsiveness.

The author conducted a cluster-randomized controlled trial across 600 paras in 29 upazilas spanning all eight administrative divisions of Bangladesh. The para was the unit of randomization. Of 600 paras initially selected, 580 entered the final study sample. Five households were sampled at baseline in each para; one household per para was set aside, leaving an analytic sample of 2,320 households, or 580 per arm. Endline data were available for 2,261 households.

Interventions and estimation

Treatment sequence

Treatment 1 provided standardized information on fees, qualifications, and waiting times, delivered verbally with a printed flyer. Treatment 2 added practical support: a facility map, clinic hours, and, where feasible, queue tokens or referral slips. Treatment 3 added encouragement and assistance to complete one public-facility visit, including guided referral and accompaniment where needed.

The primary estimates are intent-to-treat comparisons between each sequential arm and the arm immediately before it. Specifications include upazila fixed effects, the baseline outcome, and imbalanced covariates, with standard errors clustered at the para level. The paper also estimates local average treatment effects using assignment to the experience arm as an instrument for completing a public visit; the first-stage F-statistic is 47.3.

Baseline provider landscape

Measure	Informal providers	Formal providers
Share of visits	73.2%	26.8%
Mean consultation fee	40.6 taka	195.1 taka
Mean travel time	21.4 minutes	60.2 minutes
Mean satisfaction (1–7)	5.92	6.06

Informal providers were cheaper, closer, and nearly as highly rated. These observed conveniences and relational advantages make provider loyalty a meaningful economic object rather than a simple information deficit.

Results

Awareness without adoption

Information provision increased awareness that public healthcare exists by 33 percentage points, the ability to name a public facility by 38 points, and knowledge of where to access care by 37 points. Friction reduction added no significant awareness effects, while the experience arm added 6–7 points on narrower measures. Yet information alone did not change completed public visits.

Adoption effects

Incremental intervention	Attempted visit	Completed visit	Switched from informal
Information	+2 pp (n.s.)	+1 pp (n.s.)	+1 pp (n.s.)
Friction reduction	+4 pp	+3 pp (n.s.)	+2 pp (n.s.)
Experience encouragement	+25 pp***	+18 pp***	+14 pp***

Households in the experience arm were 25 percentage points more likely to attempt a public visit and 18 percentage points more likely to complete one. Among control households that completed a public visit, 9% used formal care again within two months, compared with 31% of experience-arm completers. That repeat-use comparison is descriptive rather than causal, but it is consistent with belief updating after a first visit.

Health expenditures

The control group spent 1,893 taka per month on health, or 12.1% of monthly household income—above the 10% catastrophic-expenditure threshold. Medicine spending accounted for 60% of the total. Information and friction reduction had no statistically significant expenditure effects. Experience encouragement reduced total spending by 183.4 taka and medicine spending by 142.6 taka in intent-to-treat estimates.

Effect	Total health spending	Medicine spending
Experience encouragement (ITT)	–183.4 taka***	–142.6 taka***
Induced public use (2SLS LATE)	–601.2 taka***	–467.8 taka***
Control mean	1,893 taka	1,142 taka

The 2SLS estimates imply that households induced to use public care reduced total spending by 601 taka, or 32% of the control mean, and medicine spending by 468 taka, or 41%. The paper interprets the pattern as substitution away from informal providers whose revenue incentives may encourage excess medicine use. Results are robust to winsorizing expenditure outcomes at the 90th and 95th percentiles.

Discussion: not knowing versus not wanting

The experiment separates a failure of message delivery from a deeper barrier. Thirty-three percentage points more households knew that free public care existed, but they did not change providers. The findings are consistent with households placing substantial value on familiarity, convenience, geographic proximity, social trust, and prior experience with existing informal providers.

Healthcare is a credence good because patients may not be able to verify treatment appropriateness even after consumption. It is also an experience good because patients can learn about waiting time, responsiveness, thoroughness, and respectful treatment through direct use. Information about provider qualifications does not resolve uncertainty over what an unfamiliar consultation will feel like. A first successful public consultation substitutes direct evidence for a prior formed by years of reported absenteeism, stockouts, and impersonal treatment.

The mechanism is reinforced by expenditure effects. A 41% reduction in medicine spending among induced public-care users is consistent with less exposure to informal drug sellers who face incentives to overprescribe. The finding therefore connects household financial protection to the broader public-health problem of antimicrobial resistance.

Policy implications

Awareness campaigns can achieve their proximate goal while failing to change utilization. For households already embedded in informal provider networks, experience facilitation—priority tokens, guided referrals, or accompanied first visits—may generate larger returns than information alone. Bangladesh's government community-health infrastructure and BRAC's health-worker network provide potential delivery platforms, although coverage and institutional arrangements differ across places.

Scaling is not purely a demand-side problem. If facilitated visits create congestion or expose households to inconsistent quality, the very experience intended to build trust may destroy it. Durable gains require adequate staffing, medicine availability, and service quality on second and third visits, not just a well-managed first contact.

Limitations and boundary conditions

Several limitations qualify the estimates. Queue tokens and referral slips may operate as commitment devices or trust signals rather than pure friction reduction, limiting clean separation of the second and third treatment mechanisms. Public-care use is self-reported, so social-desirability or Hawthorne effects could inflate utilization. The study does not observe administrative validation of visits or direct measures of clinical quality and health outcomes.

The follow-up spans only two months, so the experiment cannot establish whether provider switching persists for six months or two years. A positive facilitated first visit may not survive a later encounter with a different doctor, a longer wait, or a stockout. The sample is restricted to households with a child under five, a population with frequent health needs; households without young children, elderly populations managing chronic conditions, and more remote households may respond differently.

Supply-side capacity is the most important scaling constraint. The research intervention used accompaniment and priority arrangements that may be difficult to replicate without dedicated health-worker capacity. If induced demand congests already understaffed facilities, service quality could deteriorate. Seasonality in baseline data collection and imperfect reach to the household's primary care decision-maker are additional threats. Finally, the trial was registered retrospectively with the AEA RCT Registry after data collection, although the paper states that main outcomes and specifications were defined before collection.

Conclusion

Modest interventions that facilitate a single positive experience with public healthcare can produce meaningful changes in provider choice and health expenditure in rural Bangladesh. Information raises awareness but is insufficient when households remain satisfied with informal care; the relevant barrier is not simply ignorance but experience. Experience-based interventions can update beliefs, reduce reliance on overprescribing informal providers, and lower out-of-pocket spending by an economically meaningful amount. The results have direct implications for health systems in LMICs where informal providers dominate primary care and antimicrobial resistance is a growing concern.

Source: Khan, M. A. H. (2026), Health Policy and Planning, czag109. https://doi.org/10.1093/heapol/czag109. This excerpt omits references and appendices; consult the original article for full citations, tables, and supplementary material.

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Distributional Effects of Market-Oriented Reforms in Uzbekistan

A Macro-Micro Simulation Approach

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Policy Research Working Paper 11482

Abstract

Uzbekistan is advancing a comprehensive package of market-oriented reforms as part of its World Trade Organization accession strategy, including the abolition of monopolies, trade liberalization *(removal of a range of export-related subsidies and tax incentives)*, the elimination of domestic producer preferences in public procurement, and an anticipated rise in foreign direct investment. This paper evaluates the macroeconomic and distributional impacts of reforms using an integrated computable general equilibrium–microsimulation framework that combines the World Bank's MANAGE-WB model with household-level data from the 2024 Uzbekistan Household Budget Survey. The results show that the coordinated reform package generates substantial efficiency gains, raising annual gross domestic product growth by approximately 1.0 percentage point above a 5.8 percent baseline between 2025 and 2040, with an additional 0.4 percentage point gain when complemented by a foreign direct investment boost. Consumer prices fall cumulatively by up to 19.5 percent, and welfare improvements are driven primarily through enhanced purchasing power rather than nominal wage increases. The abolition of monopolies emerges as the dominant macroeconomic driver, while trade liberalization sustains long-term productivity growth and the removal of procurement preferences plays a complementary role. These gains are uneven across groups including gender, sectors, and location, and gender wage gaps remain largely unchanged. The findings underscore the importance of complementarities between competition policy, trade liberalization, and foreign direct investment promotion, as well as the need for targeted human capital, labor mobility, and social protection policies to ensure that reforms translate into broadly inclusive outcomes.

This paper is a product of the Fiscal Policy and Growth Global Department. It is part of a larger effort by the World Bank to provide open access to its research and make a contribution to development policy discussions around the world. Policy Research Working Papers are also posted on the Web at http://www.worldbank.org/prwp. The authors may be contacted at gacevedo@worldbank.org. A verified reproducibility package for this paper is available at http://reproducibility.worldbank.org, click **here** for direct access.



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Produced by the Research Support Team

1. Introduction

Uzbekistan Market Reforms and the Need for Distributional Impact Analysis.

Over the past two decades, Uzbekistan has undergone a gradual but significant economic transformation, transitioning from a centrally planned system toward a more market-oriented economy. The reform momentum intensified during the mid-2010s, as policy efforts focused on liberalizing trade and prices, strengthening private sector participation, and modernizing public institutions. While these reforms have contributed to macroeconomic stability and sustained economic growth, they have also introduced new distributional challenges and adjustment pressures across sectors and population groups.

The Presidential Decree No. UP-85 (3 June 2024) accelerates Uzbekistan's market reforms and WTO alignment by removing monopolies and exclusive rights, phasing out subsidies and preferential treatment, liberalizing trade and pricing systems, and strengthening competition and transparency. It introduces licensing and regulatory reforms; aligns taxes, customs, and procurement rules with WTO standards; and sets a clear timeline (2025–2027) for implementation to deepen global integration and improve market efficiency.

Experiences from transition and developing economies suggest that structural reforms often reshape labor markets, alter relative prices, and trigger sectoral reallocation. These changes affect household welfare through multiple channels, including employment opportunities, wage dynamics, and consumption costs. Although such reforms typically enhance economic efficiency in the long run, their short- and medium-term impacts can vary significantly across households and regions, reflecting differences in institutional capacity, labor mobility, and access to economic opportunities.

A growing empirical literature has examined the welfare and structural effects of market-oriented reforms across developing and transition economies. Evidence from trade liberalization episodes shows that reforms can stimulate productivity growth and poverty reduction, although adjustment costs are often unevenly distributed. For example, Winters, McCulloch, and McKay (2004) find that trade liberalization is generally associated with long-term poverty reduction, while Topalova (2010) documents that regions with higher exposure to tariff reductions in India experienced slower poverty reduction due to limited labor mobility. Similarly, Vadila and Resosudarmo (2020) show that tariff reductions in Indonesia contributed to reductions in income inequality, highlighting the importance of sectoral exposure to trade reform.

Research on competition and market structure reforms also emphasizes heterogeneous impacts across firms and workers. Recent firm-level evidence from Latin America shows that stronger competition policies can increase manufacturing productivity, particularly among larger and more adaptable firms, although smaller firms may face higher adjustment costs (Wong and Petreski, 2025). In parallel, studies examining subsidy reforms illustrate that efficiency gains often depend on the presence of compensatory social policies. Cross-country analysis of subsidy reform episodes suggests that gradual implementation and targeted compensation mechanisms are critical to mitigate welfare losses among vulnerable groups (IMF, 2013; Muktar, 2023).

Foreign direct investment has also been identified as an important transmission channel linking structural reforms to productivity growth and technological diffusion. While FDI can enhance long-term growth prospects, its benefits depend on human capital accumulation and financial sector development (Borensztein, De Gregorio, and Lee, 1998; Osei and Kim, 2020). Together, these findings underscore that the distributional consequences of reforms depend not only on policy design but also on country-specific structural conditions.

These lessons are particularly relevant for Uzbekistan given the country's ongoing reform agenda and demographic structure. Uzbekistan combines a relatively young and expanding labor force with persistent regional disparities in economic opportunities and productivity levels. Structural reforms aimed at liberalizing markets and strengthening private sector activity have the potential to accelerate growth and employment creation, but they may also generate differentiated welfare effects across income groups, sectors, and geographic areas. Moreover, increasing integration into global markets exposes households to new sources of income diversification while simultaneously increasing exposure to external shocks. Understanding how structural reforms translate into poverty reduction and shared prosperity is therefore essential for informing Uzbekistan's development strategy.

The World Bank has analyzed the effects of WTO accession for Uzbekistan using a computable general equilibrium model, finding substantial economy-wide gains from accession. The analysis suggests that successful integration into the WTO could increase GDP by around 10 percent over the medium to long term, while reducing poverty rates by approximately 1.6 percentage points, reflecting improvements in trade efficiency, resource allocation, and market integration (World Bank, 2024).

This paper evaluates the reforms in Presidential Decree No. UP-85 (June 2024), which provides an entry point toward a more comprehensive reform under the WTO accession. It focuses on the three core pillars of the Decree, namely (1) ending monopolies, (2) trade liberalization, and (3) removing domestic producer preferences, and additionally simulates a fourth scenario combining these interventions with an increase in foreign direct investment (FDI).

To model these scenarios, we use MANAGE and a microsimulation approach to evaluate the distributional impacts of structural reforms in Uzbekistan. By linking macroeconomic adjustments to micro-level household outcomes, the analysis provides a comprehensive assessment of both efficiency and equity implications of reform policies.

The remainder of the paper is organized as follows. Sections 2 through 5 provide an overview of the scenarios, methodology, results, and conclusions. The annexes include a literature review of the distributional impacts of each reform analyzed; an examination of advanced topics in CGE modeling; and further results from the MANAGE and microsimulation analysis.

2. Scenarios for Simulation

Uzbekistan is implementing a broad set of market-oriented reforms under Presidential Decree No. UP-85 as part of its economic transformation and WTO accession agenda. The Decree package centers on three core pillars: (i) elimination of monopolies and exclusive rights to strengthen competition, (ii) advancing trade liberalization through the removal of export subsidies and preferential tax treatments, and (iii) eliminating preferential treatment for domestic producers, particularly in public procurement. Together, these measures aim to level the playing field, improve resource allocation, and integrate Uzbekistan more fully into global markets.

Elimination of Monopolies

This scenario aims to simulate the abolition of exclusive rights held by state-owned enterprises across key economic sectors. According to Clause 1 of the Decree, a wide range of monopolistic privileges will be revoked between 2025 and 2026, a move aimed at stimulating private sector activity and aligning with WTO principles of market access and fair competition.

Key industries that are being selected for increased privatization include the metal, food, chemical, and energy sectors. First, Uzmetkombinat and Uzvtortsvetmet will lose their exclusive rights to procure ferrous and non-ferrous metal scrap. Second, Uztrade will no longer maintain its consignment rights over

flour, wheat, and goods produced by small farms. Third, Uzkimyoimpeks will no longer hold exclusive rights to export chemical products or to import the equipment and materials needed for production. Finally, monopolies held by UzGasTrade and Uzenergosotish will be dissolved, ending centralized control over natural gas and electricity trade.

Trade Liberalization

This scenario represents the removal of a range of export-related subsidies and tax incentives that currently distort Uzbekistan's trade environment. Under Clauses 4 and 8 of the Decree, the government may remove benefits like income tax refunds, property tax discounts, and transportation reimbursements for exporters. For instance, a current 50% reduction in income and property tax rates will be withdrawn from firms in the electrical sector. The 0% corporate income tax on all exports of goods and services will also be removed. Finally, current reimbursements for transportation costs, which can be up 50% for general exporters and 70% for textiles sent to Europe, Türkiye, the Arab Republic of Egypt, and Morocco will be removed as well.

These changes are critical for aligning with the WTO's Agreement on Subsidies and Countervailing Measures. They will signal a shift toward a trade environment that promotes competitiveness through productivity and efficiency, rather than fiscal incentives. As a result, Uzbekistan's export sector is expected to become more resilient and less reliant on state-provided support.

Elimination of Domestic Producer Preferences

This scenario targets preferential treatment for domestic producers in public procurement processes. According to Clause 5 of the Decree, starting from July 2026, Uzbekistan will abolish longstanding price advantages of 15%-20% granted to local manufacturers. This includes a 15% preference for general domestic manufacturers and a 20% preference for producers of electrotechnical and electric household appliances.

It aims at leveling the playing field and ensuring that procurement decisions are based on price and quality rather than origin. It supports WTO norms regarding non-discrimination and competitive neutrality: while it may increase competition for domestic firms, it also provides an incentive for them to innovate and become globally competitive.

3. Methodology

The analysis employs an integrated macro-micro modeling approach for ex-ante policy analysis. The framework combines a computable general equilibrium model with microsimulation, allowing for the evaluation of both economy-wide impacts and distributional effects across households.

This approach is particularly well suited for analyzing structural reforms, as it links changes in prices, wages, and sectoral employment generated at the macroeconomic level to household income, consumption, poverty, and inequality outcomes. Rather than producing forecasts, the framework is used to simulate counterfactual scenarios that address how the economy and different population groups would respond to specific policy interventions.

4. MANAGE

The macroeconomic analysis relies on the World Bank MANAGE-WB model, a recursive dynamic computable general equilibrium model designed to assess the economy-wide effects of structural and policy reforms. MANAGE-WB has a strong academic foundation and has been peer reviewed both

internally and externally (Beyene et al., 2025),[2] and it has been widely used in policy applications such as Country Growth and Development Reports, Country Economic Memorandums, Country Climate and Development Reports, Development Policy Operations/Financing and sector-focused analytical work.

The model solves for equilibrium on an annual basis, ensuring simultaneous clearing of all goods and factor markets in each period. Production activities are represented by firms that minimize costs under constant returns to scale, using nested constant elasticity of substitution technologies that allow for substitution between intermediate inputs, labor, and capital. This structure captures both input complementarity and substitution patterns across sectors.

Interactions between domestic and international markets are modeled using standard CGE assumptions that distinguish between domestic and imported goods and between domestic sales and exports. Import demand follows the Armington assumption, under which imported and domestically produced goods are imperfect substitutes, while the allocation of domestic production between export and domestic markets is governed by a constant elasticity of transformation function. Uzbekistan is treated as a small open economy, with world prices taken as exogenous.

Prices in the model explicitly reflect policy distortions and transaction costs. MANAGE-WB distinguishes between basic, producer, and market prices by incorporating taxes, subsidies, and trade and transport margins, allowing the analysis of both direct and indirect effects of market interventions. An extended version of the model allows for non-competitive pricing behavior through sector-specific mark-ups, which is particularly relevant for analyzing reforms that reduce market power in monopolistic sectors.[3]

Income generation and institutional accounts are modeled in a way that reflects key features of Uzbekistan's economy. Labor and capital income accrue primarily to households, with pass-through mechanisms to enterprises, which may retain earnings or transfer income to the government, reflecting the presence of partially state-owned firms. Government revenues include direct and indirect taxes, factor income from enterprise ownership, and transfers, while household consumption behavior is modeled using a constant difference in elasticity demand system consistent with the Global Trade Analysis Project model.

Labor and capital dynamics are incorporated through explicit market and accumulation mechanisms. The labor market allows for an upward-sloping labor supply curve and limited mobility across sectors and segments, while capital accumulation evolves endogenously based on aggregate investment, depreciation, and non-productive investment shares. Population and labor force growth are exogenous and typically aligned with working-age population projections.

Data, Parameters and Model Calibration

The first crucial step in implementing the MANAGE-WB model involves the detailed preparation of the necessary macroeconomic data and exogeneous parameters for model calibration. At the heart of this process lies the construction of a Social Accounting Matrix (SAM) from an Input-Output Table (IOT) or Supply and Use Table (SUT) specific to Uzbekistan's economy. The updated 2024 SAM for Uzbekistan 2024 systematically records the economic transactions between industries, households, firms, government, and external sectors, reflecting the inter-sectoral linkages fundamental to the economy. The SAM consists of 102 accounts, which include 38 activities generating 33 commodities, 6 labor segments, 3 capital accounts, 10 household categories, and 5 <4?> types of tax instruments: sales tax, export tax, tariff, direct tax.

[2] https://openknowledge.worldbank.org/entities/publication/5ab4e156-dace-4b5d-9fc8-710472d3e6f4.
[3] See annex 4 for an overview of how markups are considered in MANAGE.

The SAM is supplemented by structural parameters such as production elasticities (often modeled using Cobb-Douglas or CES production functions), substitution elasticities between imports and domestically produced goods, and consumer preference parameters using Global Trade Analysis Project (GTAP) database; and they are carefully adjusted so the model accurately replicates observed economic outcomes. Furthermore, labor market data disaggregated by skill level, education, and sector of employment are meticulously integrated to enable detailed labor market analysis, including shifts in labor demand and wages.

Scenario Specification and Definition of Policy Shocks

The second step entails specifying detailed Business As Usual (BAU) and Reforms scenarios clearly aligned with Uzbekistan's planned market liberalization initiatives. Using MANAGE-WB's flexible framework, the reference scenario (BAU) i.e. the baseline equilibrium is established, consistent with the data and estimated parameters. Then each policy reform is precisely defined as a "shock" to the baseline equilibrium.

Business As Usual

This scenario serves as the reference case and aligns with the 2025 Macro Poverty Outlook (MPO) projections.[4] Under this scenario, no policy reforms are implemented.

Table 1: macroeconomic projections under BAU

Category	2025	2026	2027	2028-2040
GDP growth	5.9	5.9	5.8	5.8
Agriculture, %VA	23.1	22.6	22.2	22.2
Industry,% VA	27.2	27.4	27.6	27.6
Service,% VA	49.8	50.1	50.2	50.2
CF,%GDP	88.5	88.7	88.9	88.9
CG,%GDP	9.9	9.5	9.1	9.1
CAB,%GDP	-5.2	-4.9	-4.8	-4.8
Fiscal Balance,% GDP	-3.0	-3.0	-3.0	-3.0

Source: Authors using the 2025 Macro Poverty Outlook. For the period 2028-2040, the projected structure of economy is assumed to be as of 2027.

In this baseline scenario, the annual growth rate is anticipated to remain stable around 5.8-5.9% between 2025-2027.

Throughout the projected period, the economic structure is expected to stay stable. Since there is no information available beyond the MPO projection period, it is assumed that this stability will continue past 2027, mirroring the trends seen beforehand. As a result, the country is anticipated to maintain a steady annual growth rate of 5.8% from 2028 to 2040. Both demand and supply sides of the economy are expected to remain stable. Furthermore, the fiscal deficit is predicted to stabilize at roughly 3% of GDP. The calibration of MANAGE-WB to align with the baseline structure (Table 1) is as follows: i) on the demand side, the consumption share as a percentage of GDP is replicated by endogenously the propensity to save. This means that, within the model, households' decisions regarding how much income to consume versus save are determined by internal economic mechanisms rather than being imposed externally. By calibrating

[4] MPO(2025).

Combined scenario

We examine how various policy mixes affect outcomes by analyzing three reform combinations: first, we pair trade liberalization with monopoly reform, since both target price distortions and market power; second, we consider combining all three policy reforms and finally we combine the full package with an exogeneous increase in Foreign Direct Investment (FDI). According to the Presidential Decree, the country anticipates increased foreign direct investment because of ongoing policy reforms (market reforms, trade liberalization), with annual growth projected at 7% from 2026 through 2030. FDI increase will boost total investment, accelerating economic growth.

MANAGE Results

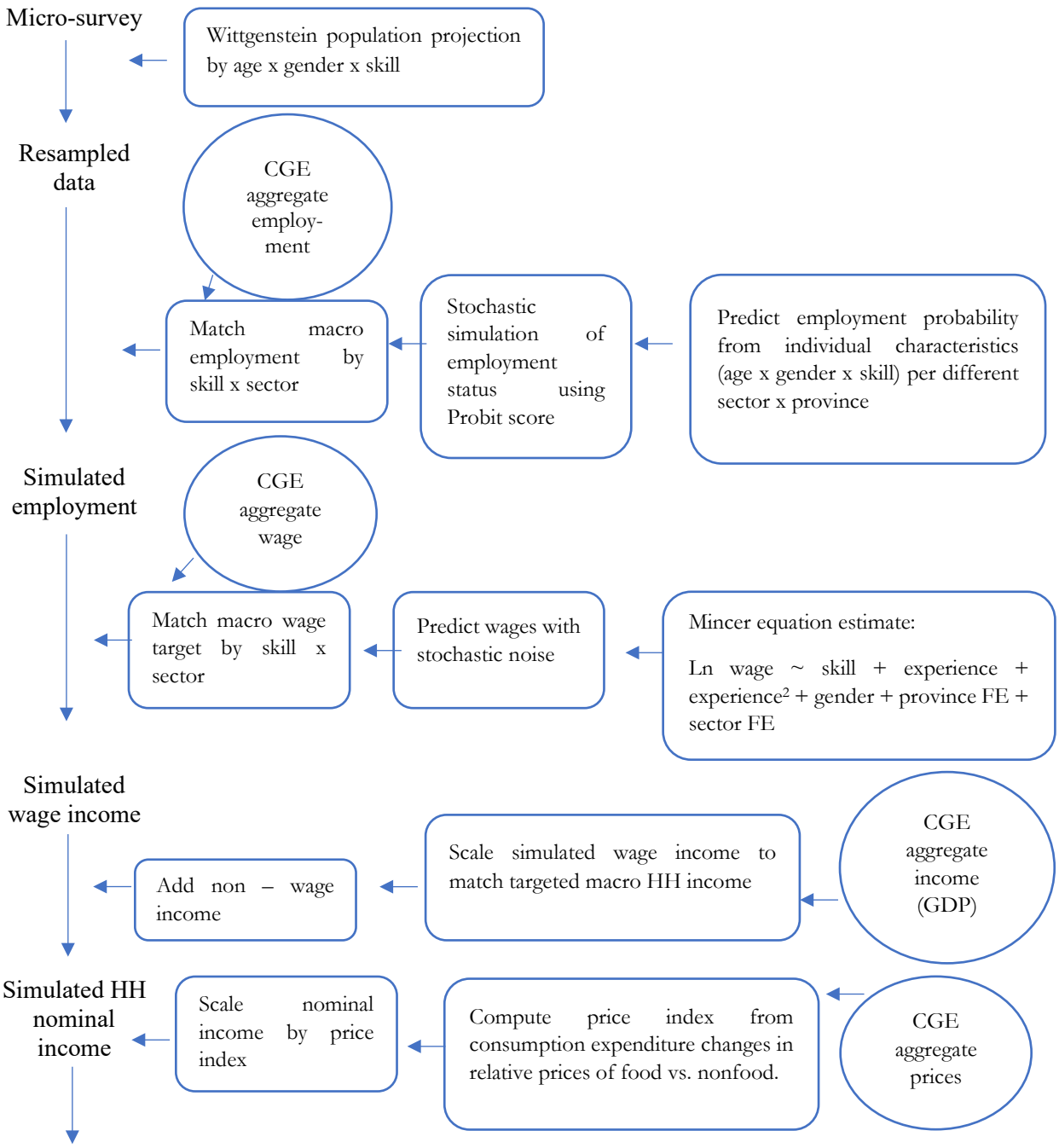
The impact of the reforms is measured by comparing the reference scenario (Business as Usual) with counterfactuals.

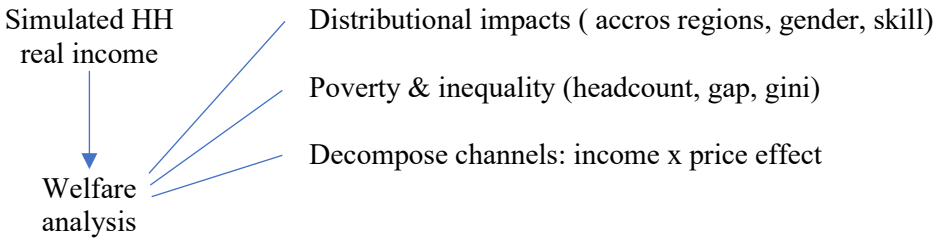
Market reforms are expected to boost competition and foster economic growth. The annual GDP growth rate is expected to be 1.0 percentage points higher with policy reforms, compared to the baseline rate of 5.8% between 2025 and 2040 (see the Tables in Annex 4). This outcome primarily stems from market reforms increasing competition, which enables more efficient allocation of resources. GDP change is mainly driven by manufacturing in industry and services sectors on the supply side, while exports and public and private consumption drive demand. Our findings indicate that growth is primarily driven by efficiency rather than investment, as shown by rising GDP alongside declining total investment relative to the baseline. This outcome is largely due to the elimination of markups under monopoly conditions. Specifically, removing markups (and thereby deadweight loss) in manufactured goods reduces production costs for firms using these goods as input, facilitating the reallocation of production factors and increasing output. However, given that the model is savings-driven, the overall savings pool may contract following the reform. This occurs because the corporate rents previously safeguarded by markups are removed, resulting in lower capital income distributed to wealthier households, who typically exhibit a higher propensity to save. The rise in exports is driven by long-term productivity gains from trade liberalization as explained in the text and greater economic competitiveness due to lower prices from mark-up removal.

Eliminating market power in manufacturing sectors leads to the most significant effects. Our analysis shows that removing monopolies from these targeted sectors alone could boost the GDP growth rate by 0.8 percentage points above the baseline of 5.8%. This is because the reform lowers market prices in those sectors, which improves competitiveness and stimulates greater final and intermediate demand. The findings also indicate that while trade reforms may yield modest short-term GDP growth (0.1 ppt above the baseline), they are projected to substantially foster long-term economic growth through enhanced efficiency (0.3 ppt). The limited short-term impact arises because, in the absence of export tax incentives, domestic producers face reduced competitiveness internationally, which leads to lower output. Consequently, these producers shift their focus toward the domestic market, benefiting local consumers. In the long term, however, domestic firms can adapt by embracing innovation to enhance their competitiveness against foreign counterparts. Without government support, companies are further motivated to reduce costs, improve quality, and identify new markets to maintain their competitive position. The elimination of preferences for domestic producers appears to have a small effect.

On average, the combined reform is expected to lower the consumer price index by 19.55% from 2025 to 2040 relative to the baseline scenario (refer to Table A4.2 in the appendix). This reflects a cumulative reduction in price throughout the period from 2025 to 2040. The greatest impact will come from trade reforms, reducing the index by 14.7%, followed by the removal of the monopoly at 7.3%. The combined effect is smaller than the sum of the individual impacts because of general-equilibrium

Figure 4 Framework





Reweighting of Household Data

The first step involves the reweighting of household survey data to match projected population characteristics under a future macroeconomic scenario. This is essential for ensuring demographic consistency between micro-level data and macroeconomic assumptions. The reweighting procedure adjusts the sampling weights of households such that the distribution of key demographic variables such as age, gender, and educational attainment, aligns with population projections from Wittgenstein Population Projection for different age groups, genders and education levels.

Employment Status Simulation

The second step of the analysis models how employment adjusts across sectors and skill groups in response to macroeconomic reforms or shocks. Employment status is modeled using a logit specification estimated on the original data, with employment probabilities depending on individual characteristics such as age, skill level, gender, and province of residence. The estimated coefficients capture systematic differences in employability and are interpreted as structural relationships that remain stable across policy scenarios.

These coefficients are applied to the simulated population to generate individual employment probabilities. To align the outcomes with the macroeconomic target, predicted probabilities are calibrated to the macro employment target using a uniform intercept adjustment. Employment status is then assigned stochastically by comparing calibrated probabilities to fixed random draws, ensuring realistic individual-level variation and comparability across counterfactual scenarios. Any residual deviation from the macro employment target is corrected ex-post through minimal adjustments for individuals at the margin.

Sectoral Reallocation

Sectoral allocation among employed individuals is modeled using a multinomial logit estimated on the original survey data, relating sector choice to age, skill level, sex, and province. The estimated coefficients capture relative sectoral affinities conditional on observable characteristics and are used to rank individuals' suitability across sectors.

After employment status has been simulated, individuals are classified into labor market flow categories based on their pre and post simulation employment status, including employed to employed workers, employed to unemployed workers, unemployed to employed workers, and unemployed to unemployed individuals.

Reallocation is performed separately by skill group to match skill-specific employment targets. Employed individuals are retained in their initial sector if their sector affinity is sufficiently strong relative to the sector's employment target; otherwise, they are released and become eligible for reassignment. Released stayers and new hires form a common pool competing for remaining vacancies, which are filled again

according to sector-specific affinity ranking. This iterative procedure continues across sectors and skill groups until all sectoral targets are met.

Wage Assignment

Following sectoral reallocation, wages are assigned to all simulated employed individuals using a standard Mincer estimation that associates individual earnings to sector of employment, province, labor market experience, experience squared, gender, and other relevant observable characteristics. The estimated coefficients capture systematic wage differentials associated with sectoral affiliation, geographic location, and individual characteristics, as well as the concave relationship between earnings and labor market experience. The estimated earnings function is then applied to simulated employed individuals using their post-reallocation characteristics, including their simulated sectors. Predicted wages therefore reflect both individual heterogeneity and sector-specific wage premia, ensuring consistency with baseline earnings patterns while allowing adjustment to changes in sectoral employment composition.

Relative Price Effect and Nonlabor Income Adjustment

The final component of the micro-simulation is the incorporation of relative price changes implied by the CGE model. To account for relative changes in the cost of consumption, the household nominal income is deflated both temporally using total CPI change over time and basket-specific using a Laspeyres style deflator computed from food and non-food price changes.

Finally, as labor income has met its macro target, total household income is calibrated to the macro per capita GDP (household expenditure) by adjusting the non-labor income residually. In particular, the calibrated mean non-labor income is computed as the difference between per capita macro income and simulated per capita wage. The ratio between unadjusted and calibrated mean non-labor income forms a scale factor, which then can be applied to each household's non-labor income.

Here, we construct a synthetic multi-year survey panel that is fully consistent with macroeconomic aggregates, specifically employment, wages, prices, and household income, while preserving individual-level heterogeneity. This integrated framework enables us to derive key outcome variables, including employment, real wages, and household income, and trace their evolution over time. These variables form the basis of our distributional analysis, which systematically examines how reform impacts vary across population groups, with results disaggregated by gender, skill level, and region.[7]

[7] When integrating the results from the CGE (Computable General Equilibrium) model with the microsimulation framework, we use household income per capita, carefully accounting for both labor and non-labor income components. During this process, we observed an uneven and unrealistic distribution of non-labor income across varying income levels within the micro-survey data. Our analysis shows that households with lower incomes get a larger share of non-labor income than those with higher incomes. This pattern is inconsistent with established economic understanding, which suggests that poorer households typically depend primarily on labor income, whereas wealthier households diversify their earnings through capital gains and other non-labor sources. To prevent bias in poverty analysis, we focus solely on labor income heterogeneity and preserve original nonlabor income distribution from the micro-survey without reconciling macro- and micro-level data. By doing so, we intentionally exclude the problematic redistribution of non-labor income, allowing the inherent heterogeneity of the labor market and the structural characteristics of household expenditure patterns to drive the distributional analysis. This approach provides

5.2. Microsimulation Results

Growth, Poverty and Mechanism

- ***Adjusted Per-Capita Income by Scenario***

Figure 5 reports the percent change in adjusted per-capita income relative to the baseline, so all trajectories start at zero and reflect cumulative deviations generated by each reform scenario. During the first periods, income responses remain modest across all scenarios, indicating that household welfare reacts gradually as sectoral reallocation and price adjustments unfold rather than through immediate shocks.

Over time, combined reform scenarios separate visibly from individual reforms. The package combining trade liberalization and monopoly abolition follows a trajectory that closely tracks the full reform scenario, suggesting that most income gains originate from the interaction between increased openness and stronger domestic competition.

Among individual reforms, abolition of monopolies is the strongest driver of adjusted income, rising steadily over the projection horizon and approaching the trajectory of the full reform package. Trade liberalization on its own produces only modest gains, ending well below the monopoly abolition path and indicating that openness delivers limited income effects unless paired with stronger domestic competition. Procurement reform shows almost no deviation from the baseline, reinforcing the idea that changes in public purchasing rules have weak aggregate income effects when implemented in isolation.

The combined scenario, including all reforms plus FDI, shows strong income gains and ultimately surpasses the "All" scenario toward the end of the period. This suggests that the inclusion of FDI contributes additional aggregate income gains over the medium to long term, even if its effects are not always immediate (Annex 6).

Figure 5: Deflated per-capita income by scenario

a more credible assessment of poverty and income distribution, ensuring that the results are not distorted by unrealistic assumptions about non-labor income allocation.

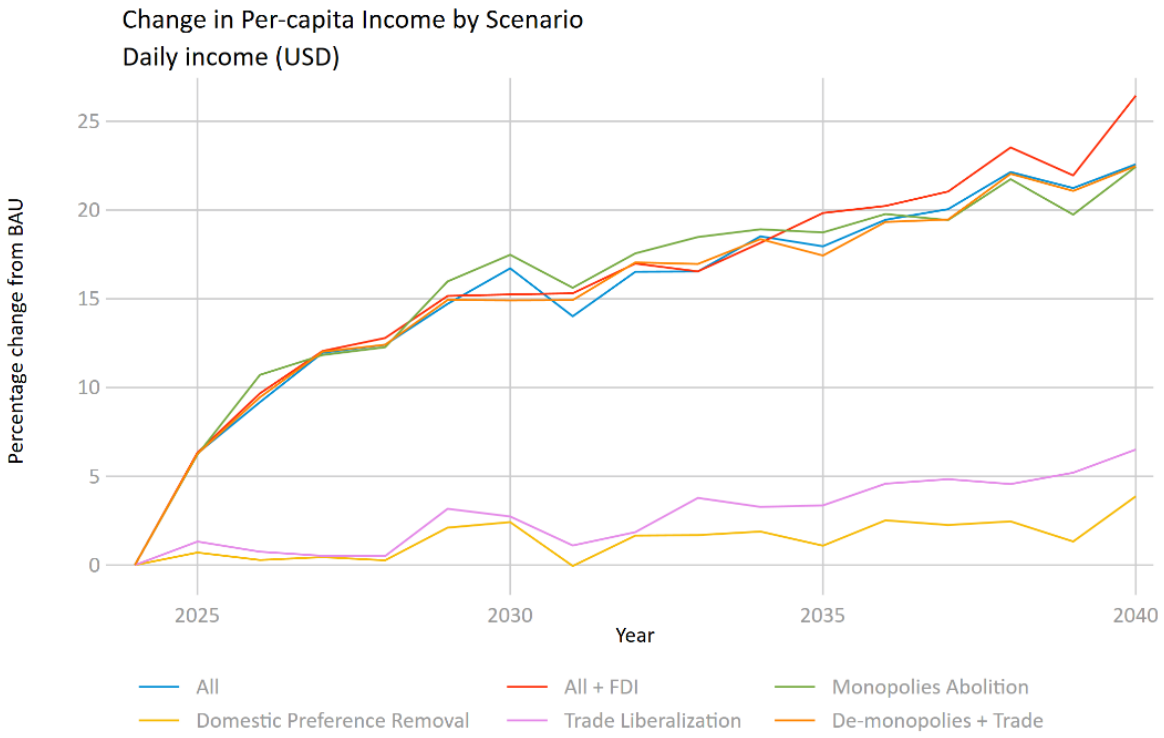


Figure 6 shows changes in the poverty rate across different scenarios using the USD 8.3 (UMIC) poverty line. Taken together, Figures 5 and 6 indicate that welfare gains do not translate uniformly into poverty reduction. While comprehensive reform scenarios, particularly those combining multiple measures, tend to reduce poverty more consistently, partial reforms such as trade liberalization or the removal of domestic preferences raise poverty relative to the baseline throughout the projection horizon, ending roughly 0.7 to 1.5 percentage points above it. This suggests that the magnitude and persistence of welfare improvements depend not only on income gains, but also on how reforms jointly shape prices, labor markets, and income distribution, highlighting the importance of complementarities across policy measures.

The different results across alternative thresholds (the vulnerability line, the national poverty line, and the USD 8.3 line) are reported in Annex 7.

in both locations, but the gains are modest and unevenly distributed over time, with rural areas in particular showing transitory increases before settling below the baseline.

Figure 14: Change in Poverty rate (below National Poverty Line) by location under Full Reform package

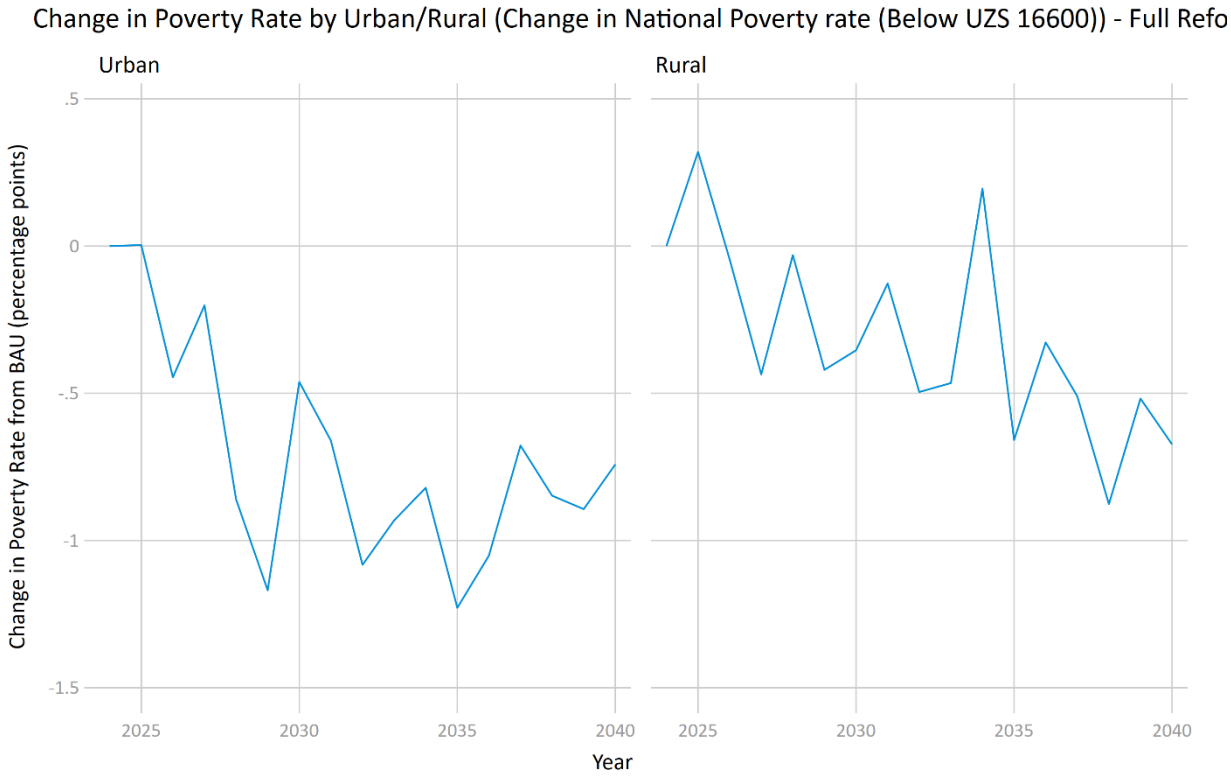


Figure 15 presents the simulated changes in the vulnerability line (population with income between UZS 16,600 and UZS 35,993). The absolute magnitudes are markedly larger than under the lower poverty thresholds, reflecting the wider segment of the population captured by the vulnerability band. The urban panel falls steadily from the start of the simulation, reaching roughly -3 to -3.5 percentage points by 2030 and oscillating around that level through 2040, where it ends close to -3.5 percentage points. The rural panel declines more sharply and continues to drift downward with visible volatility, reaching a trough near −6.2 percentage points around 2036 and ending close to -6 percentage points by 2040. The reform package therefore reduces vulnerability in both locations, but the rural reduction is roughly twice as large as the urban one, reversing the pattern observed at the lower poverty lines and consistent with a higher initial concentration of near-poor households in rural areas.

Figure 15: Change in Poverty rate (below vulnerability line) by location under Full Reform package

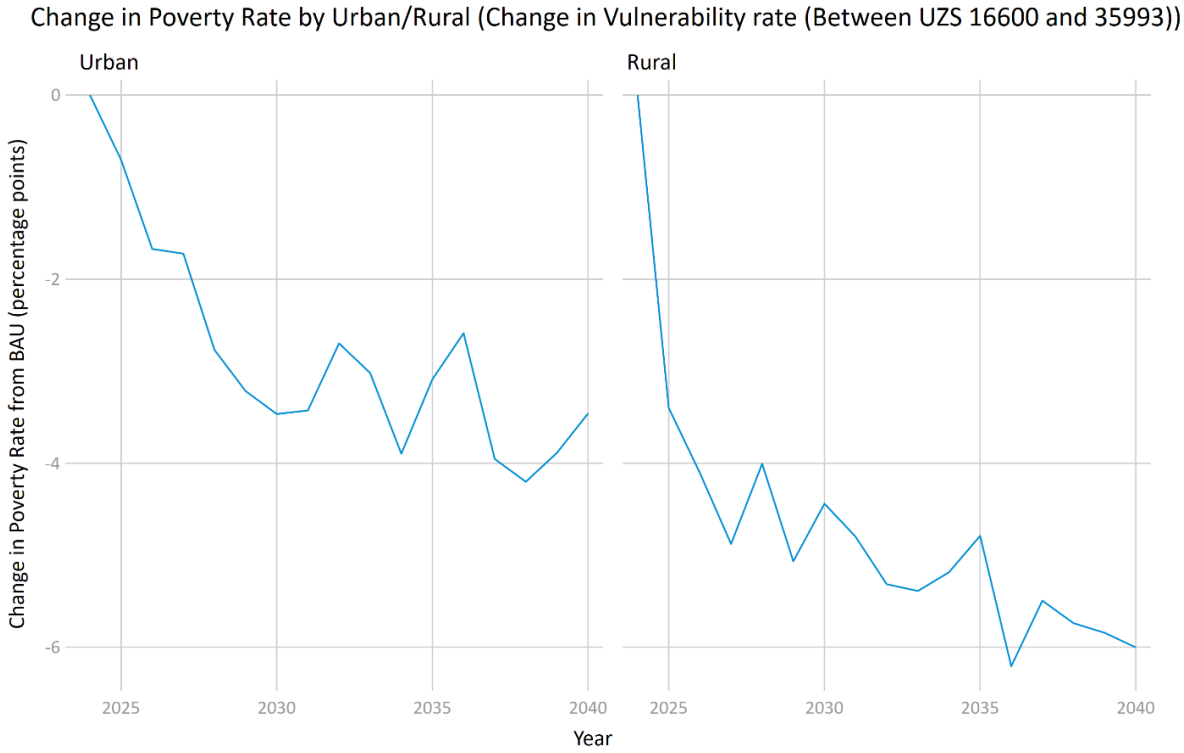


Figure 16 plots the change in the vulnerability rate (population with income between UZS 16,600 and UZS 35,993) by scenario. Because the vulnerability band captures a wider segment of the population than either of the lower poverty lines, the absolute reductions across scenarios are noticeably larger. Among individual reforms, monopoly abolition is again the strongest driver, reducing vulnerability by roughly 5.2 percentage points by 2040, while the combined trade-and-monopoly path tracks the full reform package closely throughout the horizon. The full reform package augmented with an FDI boosting target ("All + FDI") delivers the largest reduction, near -5.7 percentage points and pulling away from the other scenarios in the later years. By contrast, trade liberalization in isolation produces only a modest decline of about 1 to 1.5 percentage points, and the removal of domestic producer preferences hovers near zero throughout the horizon, briefly turning slightly positive around 2035. Compared with the lower poverty thresholds, the decomposition confirms the broader result that monopoly abolition is the dominant channel for vulnerability reduction, with FDI providing additional impetus in the long run, while partial liberalization measures on their own have limited transmission to vulnerability outcomes.

Figure 16: Change in vulnerability rate by scenario

Natural Disasters and Growth – Going beyond the Averages[*]

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JEL Classification: O11, O40, Q54

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Policy Research Working Paper 4980

Abstract

There has been a steady increase in the occurrence of natural disasters. Yet their effect on economic growth remains unclear, with some studies reporting negative, and others indicating no, or even positive effects. These seemingly contradictory findings can be reconciled by exploring the effects of natural disasters on growth separately by disaster and economic sector. This is consistent with the insights from traditional models of economic growth, where production depends on total factor productivity, the provision of intermediate outputs, and the capital-labor ratio, as well as the existence of important intersector linkages. Applying a dynamic Generalized Method of Moments panel estimator to a 1961–2005 cross-country panel, three major insights emerge. First, disasters affect economic growth—but not always negatively, and differently across disasters and economic sectors. Second, although moderate disasters can have a positive growth effect in some sectors, severe disasters do not. Third, growth in developing countries is more sensitive to natural disasters—more sectors are affected and the magnitudes are non-trivial.

This paper—a product of the East Asia and Pacific Social Protection Unit and the Development Research Group—is part of a larger effort to study main sources of vulnerability. Policy Research Working Papers are also posted on the Web at http://econ.worldbank.org. The authors may be contacted at nloayza@worldbank.org, and jrigolini@worldbank.org.

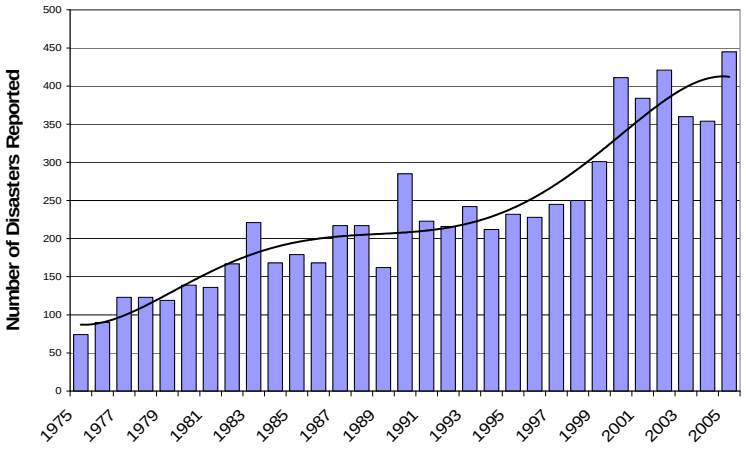
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Produced by the Research Support Team

1. Introduction

Along with climate change has come an increase in the frequency of natural disasters across the world (Figure 1). This poses an important policy challenge. Natural disasters cause tremendous human suffering. Locally, they often also yield substantial physical and economic damages, which may temporarily, or even permanently, jeopardize a country's overall economic development. To help policymakers gauge the benefits from disaster risk mitigation and adaptation, it is important to better understand the economic costs associated with natural disasters.

Figure 1: Trends in Natural Disasters, 1975-2005



Source: author's own calculations using data on natural disasters from CRED- EMDAT.

This has instigated an incipient literature on the empirical relationship between natural disasters and economic growth. As expected, several papers report a (substantive) negative effect of disasters on growth. For instance, using a cross-country sample for the period 1970-2002, Rasmussen (2004) finds that natural disasters lead to a median reduction of 2.2 percent in the same-year real GDP growth, and that they increase the current account deficit and public debt.[1] Surprisingly however, many others find no

[1] Other studies that report a negative effect include Raddatz (2007), Heger, Julca, and Paddison (2008), and most recently, Noy (2009). Based on reviews of events (as opposed to cross-country studies), Charveriat (2000), Crowards (2000), and Auffret (2003) also find that major events are associated with drops in aggregate output.

effect, or at times even a positive one. Testing the empirical validity of the predictions of the Solow model, Caselli and Malhotra (2004) fail to find a negative relationship between natural disasters and medium-term aggregate economic growth. Similarly, Albala-Bertrand (1993, Ch. 4) find no or little effect.

Jaramillo (2007) observes that the sign and magnitude of the relationship depends on the type of disaster. Skidmore and Toya (2002) consider average per capita GDP growth over 1960-1990 and find that climatic disasters are associated with higher long-run economic growth, while geologic disasters are negatively associated with growth. In analyzing long-term empirical relationships, causality considerations are however substantially complicated, as countries may have adopted (less remunerative) technologies that are less sensitive to frequent disasters.

In sum, the current empirical literature remains inconclusive about the effects of natural disasters on growth. This should not necessarily come as a surprise, as theory suggests that different types of disasters can have diverse (even opposite) effects on growth. Disasters that affect the provision of essential intermediate inputs in production, for instance, such as droughts in agriculture, should have an adverse impact on growth, but disasters that affect adversely the capital-labor ratio, such as earthquakes, can in principle have a positive impact on growth through increasing returns and high reconstruction investments. Consequently, the impact should also vary across sectors (and given their differing relative importance, also across countries): for instance, droughts are likely to significantly affect agriculture, but less so industry, while earthquakes are more likely to affect industry.

Drawing on insights on the dynamics of economic growth from the stylized Solow-Swan growth model, this paper seeks to reconcile the apparent contradictions in the current empirical literature through a more systematic recognition that different disasters affect economic sectors through different channels and that, as a result, their effects are likely to differ by the type of disaster, and also across sectors and countries, depending on their level of economic and institutional development[2].

[2] A related strand of literature demonstrates that the quality of a country's institutions, its democratic election processes, educational attainments, and openness reduce casualties and damages, and improve macroeconomic performance after the event (Kahn, 2003; Rasmussen, 2004; Toya and Skidmore, 2005; Skidmore and Toya, 2007; Noy, 2009).

The focus is on medium-term economic growth (5-year periods) thereby mitigating potential biases due to adaptation. The effects of the different natural disasters (i.e. droughts, floods, earthquakes and storms) are examined separately by economic sector (agriculture, industry, and services), each time controlling for a series of well known growth determinants. This way the paper broadens the scope of the existing literature, which has so far largely concentrated on aggregate measures of disasters and/or economic activity. This disaggregated approach also yields preliminary insights in the distributive effects of natural disasters. Through the use of the dynamic panel GMM estimator developed by Arellano and Bover (1995) and Blundell and Bond (1998) great care is taken in addressing endogeneity issues related to the potential correlation between explanatory variables and unobserved country-specific factors.

To maintain consistency with other studies, the data on natural disasters are obtained from the Emergency Disasters Database (EM-DAT) database of the Centre for Research on the Epidemiology of Disasters (CRED). The share of the population affected by a specific type of disaster over a given period of time is taken as measure of natural disaster. This way, both the *frequency* and *intensity* of the disaster are reflected. The sample spans 94 developing and developed countries over the period 1961-2005.

The empirical results are consistent with the implications from the traditional Solow-Swan model. Three major conclusions emerge. First, different disasters affect growth in different economic sectors differently and the insights obtained with over-aggregation are misleading. Second, while moderate disasters can have a positive growth effect on certain sectors, severe disasters don't. Third, growth in developing countries is more sensitive to natural disasters—more sectors are affected, the magnitudes are non-trivial, and the poor are likely to be more affected by disasters (both positively and negatively).

To motivate the disaggregated approach and facilitate the interpretation of the empirical results Section 2 proceeds by reviewing the Solow-Swan growth model. Section 3 discusses the estimation methodology and section 4 elaborates on the growth determinants and the natural disaster data used. The empirical findings are discussed in section 5. Section 6 concludes.

2. Conceptual Framework

To better understand through which channels natural disasters may affect economic growth across sectors and to better motivate the disaggregated approach, the basic elements of the Solow-Swan growth model are revisited. This well-known model has been used extensively in the past for its conceptual strength and clarity in elucidating the process that occurs in the transition to a long-run steady state. This is reflected in the medium-term economic growth variables and forms the relevant time horizon for this paper.

Consider a production function with decreasing marginal returns, constant returns to scale, three production factors, and a general productivity parameter. For simplicity, assume a production function of the Cobb-Douglas form:

$$Y = AK^{\alpha}L^{\beta}M^{1-\alpha-\beta} \qquad (2.1)$$

Where, Y is output, A represents the general productivity parameter, K is capital, L is labor, M represents materials and other intermediate inputs, and α, β, $1\text{-}\alpha\text{-}\beta$ are the corresponding factor shares (all between 0 and 1). The marginal product of each factor is positive but decreasing (with limits of ∞ and 0 as the factor approaches 0 and ∞, respectively).

The action in the Solow model is given by its dynamic equations. It is assumed that only one factor of production, capital, is accumulated purposively. A constant fraction of output is saved and invested in capital formation. Labor follows an exogenously fixed growth rate. Productivity and intermediate inputs can change arbitrarily. Thus,

$$\Delta K = sY - \delta K \qquad (2.2)$$

$$\Delta L = nL \qquad (2.3)$$

Building on these basic insights, droughts are expected to have a negative effect on agricultural growth because they entail a drastic reduction of water, a vital input in agricultural production. These negative effects likely extend to industrial growth through two mechanisms both related to the provision of raw materials and intermediate inputs. The first is by reducing the supply of agricultural products that serve as inputs to the (agro-processing) industry. The second is by hampering electricity generation, particularly where hydropower is a major source of electricity. In addition, their negative effect may be compounded by the fact that droughts affect people and workers much more than they destroy physical capital, thus increasing *k* beyond its steady state level.

Floods induce a disruption of farming, urban activities, and transportation in the areas most affected by them, negatively affecting overall productivity. When floods are severe and long lasting, the emergence of water borne diseases may further exacerbate this decline in TFP. However, when floods are localized and moderate, they could also be associated with higher growth through a variety of mechanisms. In agriculture, floods may raise growth by increasing both the supply of water for future irrigation and land productivity. They may also reflect more abundant rainfall nationwide. In industry, floods may increase growth by raising the supply of agricultural products and electric power, both important intermediate inputs for industrial production. The positive effect of floods on services growth may also come through inter-linkages with other sectors (e.g., a larger supply of inputs for commerce and retail).

Earthquakes may have a positive impact on industrial growth. Although they severely affect both workers and capital, earthquakes particularly destroy buildings, infrastructure, and factories. The capital-worker ratio is then sharply diminished, the average (and marginal) product of capital increases, and output grows as the economy enters a cycle of reconstruction. Moreover, if destroyed capital is replaced by a vintage of better quality, factor productivity increases, leading to a further push to higher growth.

Storms may have a negative effect on agricultural growth, but, if they are not severe, a positive one on industrial growth. Agricultural growth declines after storms because they destroy the seedlings and plants (or the harvest) on the fields, which are intermediate inputs in the final product. Storms also destroy considerable amounts of physical capital important in industrial production, devastating capital relatively more

than incapacitating workers. As the capital-worker ratio drops, this mechanism would suggest a growth expansion in industry.

3. Estimation Methodology

The point of departure is a standard growth regression equation designed for estimation using (cross-country, time-series) panel data:

$$y_{i,t} - y_{i,t-1} = \beta_0 y_{i,t-1} + \beta_1 CV_{i,t} + \beta_2 ND_{i,t} + \mu_t + \eta_i + \varepsilon_{i,t} \qquad (3.1)$$

Where the subscripts *i* and *t* represent country and time period, respectively; *y* is the log of output per capita, *CV* is a set of growth control variables, and *ND* represents natural disasters; μ_t and η_i denote unobserved time- and country-specific effects, respectively; and ε is the error term. The dependent variable ($y_{i,t}$-$y_{i,t-1}$) is the average rate of real output growth (i.e., the log difference of output per capita normalized by the length of the period).

The regression equation is dynamic in the sense that it includes the level of output per capita ($y_{i,t-1}$) at the start of the corresponding period in the set of explanatory variables. This poses a challenge for estimation given the presence of unobserved period- and country-specific effects. While the inclusion of period-specific dummy variables can account for the time effects, the common methods of dealing with country-specific effects (that is, within-group or difference estimators) are inappropriate when a regression is dynamic in nature.

The second challenge is that most explanatory variables are likely to be jointly endogenous with economic growth, so we need to control for the biases resulting from simultaneous or reverse causation. Although natural disasters are exogenous–and treated as such in the econometric estimation[4]—their effects would be incorrectly estimated if the endogeneity of the remaining variables in the model is ignored.

[4] The measure of external shocks (i.e. growth rate of terms of trade—see below) is also considered exogenous.

of pair-wise correlations of these variables. All data except the data on natural disasters are from World Bank's World Development Indicators, WDI, (2007).

Four dependent variables are considered. For comparison with other studies, regressions are first run using the growth rate of real per capita Gross Domestic Product (GDP) as dependent variable. Subsequently measures of the growth rate of real per capita value added in the three major sector of the economy, that is, agriculture, industry and services are used. All of them are measured as the five-year average of the log differences of per capita output (in 2000 US dollars). Per capita output is obtained by dividing the value added of each sector by the total population.

From Appendix 1 it emerges that the growth performance of different sectors has been diverse: the service sector has grown the fastest (1.83 percent per year), followed by industry (1.73%), and agriculture (0.33%). The disparity across sectoral growth performance would be consistent with the view that natural disasters have diverse effects on the different sectors of the economy.

Three groups of growth determinants are considered: 1) variables that measure transitional convergence, structural and stabilization policies, and institutions; 2) variables that proxy the role of external conditions that may affect the growth performance across countries; and 3) natural disasters, which form the subject matter of the paper. To control for transitional convergence, in each regression the corresponding initial value of output per capita (in logs) for the five-year period is used. This is crucial to test whether the initial position of the economy is important for its subsequent growth, all things equal. A negative sign would suggest that poor economies tend to catch up and grow faster than rich economies.

Similar to the cross-country growth specifications by Levine, Loayza, Beck (2000) and Dollar and Kraay (2004) the areas of education, financial development, monetary and fiscal policy, and trade openness are considered to capture the role of structural and stabilization policies, and institutions. Education is approximated by the log of the gross rate of enrollment in secondary school, which is the ratio of the number of students enrolled in secondary school to the number of persons of the corresponding age. Financial depth is measured by the ratio of private domestic credit supplied by private financial institutions to GDP. The government burden is measured as the ratio of

CRED defines a disaster as "a situation or event which overwhelms local capacity, necessitating a request to a national or international level for external assistance; an unforeseen and often sudden event that causes great damage, destruction and human suffering." For a disaster to be entered into the database, at least one of the following criteria must be fulfilled: 10 or more people reported killed; 100 or more people reported affected; declaration of a state of emergency; or call for international assistance.

CRED divides disasters according to type (for example: drought, flood, etc), and provides the dates when the disaster occurred and ended; the number of casualties (people confirmed dead and number missing and presumed dead); the number of people injured (suffering from physical injuries, trauma or an illness requiring immediate medical treatment as a direct result of a disaster), and the number of people affected. People affected are those requiring immediate assistance during a period of emergency (i.e. requiring basic survival assistance such as food, water, shelter, sanitation and immediate medical help). People reported injured or homeless are aggregated with those affected to produce the "total number of people affected".

Finally, EM-DAT also provides an estimate of "economic damage". Although "economic damage" could be a good indicator of the gravity of a disaster, it has important drawbacks both from a measurement and estimation perspective. First, CRED admits that there is no standard procedure to determine economic impact. Second, economic losses are reported for only one third of the disasters, with the proportion differing substantially across the types of disasters.[13] Third, such a measure would make the exogeneity assumption tenuous, as the amount of damage may be correlated with the growth during the period under consideration.

[13] For example, economic losses are reported for nearly 50% of all the windstorms entered in EM-DAT and 40% of the earthquakes. This is most likely due to the infrastructure damage that is directly and clearly attributable to these events. Floods are the third largest category, with losses reported for about one-third of the total events. For droughts, on the other hand, less than 25% of the events have losses reported. There may be several factors for this. In particular, CRED recognizes that droughts may only draw the international attention in terms of lives lost, with little consideration for economic costs. Droughts do not result in infrastructure or shelter damage but in heavy crop and livestock losses, therefore, most economic losses are of an indirect or secondary nature and difficult to quantify.

population, is taken as measure of disaster, or formally, $ND_{i,t} = Log\left(\sum_j \frac{Total\ Affected_{i,t,j}}{Population_{i,t}}\right)$

where j indexes the number of events that took place in country *i* during (five-year) period *t*. By considering the sum of the number of people affected per event, the measure explicitly accounts for both the *frequency* and the *intensity* of the shock, contrary to many of the measures used in the literature. To enable comparison across countries, further normalization by the total population is undertaken to correct for differences in population size.

Inspection of the distribution of the weighted sum of natural disasters shows that it is positively skewed. Consequently the log is taken to avoid that the empirical results are driven by extreme values. Not to lose too many observations (observations for which no event has been reported result in an undefined value of the log of the disaster measure), these observations are assigned a value of to -20, which is just below the lowest observation for which an event was reported.[16] Finally, natural disasters such as storms and floods often occur in tandem—Appendix 3 indicates a correlation of 0.22 between floods and storms, and a correlation of 0.24 between floods and earthquakes. To isolate the effects of each natural disaster, the four natural disaster measures are included simultaneously in the regressions.

5. Empirical Results

Table 1 presents the basic estimation results using the full sample. The results in the first two columns pertain to the growth rate of GDP per capita, while those in the last three columns pertain to per capita valued added output growth rates in agriculture, industry, and services, respectively. The same set of set of explanatory variables is included as control variables across all regressions, except that initial output corresponds to the initial valued-added of the respective sector. The Hansen specification and serial-correlation tests indicate that the null hypothesis of correct specification cannot be

[16] This number has been (arbitrarily) chosen to be low enough not to affect the distribution of the natural disaster indicator.

disasters are robust to these alternative specifications of the traditional growth control variables.

Turning to the growth effects of natural disasters, natural disasters are found not to affect GDP growth when using a combined index of natural disasters—the sign of the coefficient is positive but statistically insignificant (Col. 1). The lack of a significant effect reflects well the theoretical ambiguity and the diverging empirical findings reported in the literature to date. Indeed, when disaggregating by type of natural disaster (col 2), coefficients of contrasting signs emerge (negative for droughts and earthquakes, and positive for floods and storms). However, except for floods, they fail to be statistically significant. To better understand how the different disaster affect growth (and also poverty), further disaggregation of growth by economic activity is warranted.

In contrast to the weak effects on overall GDP growth, three types of natural disasters appear statistically relevant for the growth of agricultural output (Col. 3): droughts and storms carry negative coefficients, while floods a positive one. On the other hand, the effects on industrial and service output growth are rather weak for the sample of all countries. In the case of industrial growth (Col. 4) no coefficient appears to be statistically significant. For service growth, floods are the only natural disaster that carries a significant coefficient, with a positive sign that starts to become robust.

When looking at the sample of developing countries only (Table 2), the growth effects of natural disasters are stronger in significance and, in some cases, also magnitude. When neither GDP growth nor the index of natural disasters is disaggregated by sector or type (Table 2, Col. 1), the coefficient on natural disasters is positive but not statistically significant. As before, the results gain significance and diversity once disaggregated. When the four types of natural disasters are considered individually but jointly in the regression, both droughts and floods appear to have a significant effect on per capita GDP growth, with droughts decreasing and floods raising growth (Col. 2).

The effects on agricultural growth are given in Col. 3. As in the full sample, droughts and floods have the largest but opposite effects. . The impact of droughts is clearly negative on agricultural growth while that of floods is positive, though somewhat smaller than that for the sample as a whole. Interestingly, holding constant droughts and floods, the effect of storms is negative and significant for agricultural growth. This

lack of preparation. As discussed above, if an event produces a sharp reduction in the capital-labor ratio, it is likely to be followed by higher growth, and *vice versa*. The industrial sector further stands to receive a growth boost from the demand for capital reconstruction that follows earthquakes and storms in areas including housing, infrastructure, and manufacturing.

Lastly, the effects on the growth of service output are assessed (Col. 5). In this case, only floods carry a significant coefficient, indicating a positive effect of floods on services output growth. Given that this sector includes commerce and retailing, among other cross-cutting economic activities, services have strong links with both agriculture and industry, especially in developing countries, as suggested by the larger coefficient for the sample of developing countries than for the full sample. Therefore, the positive impact of floods may be partly the result of its beneficial impact on agricultural and industrial outputs.

Another mechanism through which service growth may be affected by natural disasters is that relief resources and activities increase the demand for service-related sectors, such as transport and communications, banking, and government. This effect will complement the effects coming through other mechanisms, adding size to the positive ones and reducing the magnitude of the negative ones. Thus, in the case of floods, the positive effect of relief activities increases the beneficial spill-over of agriculture and industry. This may also be the reason why the effect of droughts on service output growth is not statistically significant: the positive relief effect counteracts the negative spill-over effect coming from agriculture and industry. Finally, unlike industry, services tend to be less intensive in physical capital, and more intensive in telecommunication and infrastructure. As a result, it is less likely to receive growth support from a decline in its capital/labor ratio, and more likely to suffer from a decline in its total factor productivity following an earthquake or a storm.

Two further robustness tests were conducted using the sample of developing countries only. The first exercise concerns the estimation methodology and re-estimates the growth regressions using a standard least-squares (LS) methodology, rather than the more complex GMM estimator (see Table 3). In this case, the statistically significant results are a subset of those under the preferred GMM methodology. That is, there is no

Chart 2: Growth effect of a "typical" (median) natural disaster

		Effect on:			
		GDP Growth	Agricultural Growth	Industrial Growth	Service Growth
Median intensity:	Droughts	-0.606 ***	-1.071 ***	-1.029 **	-0.127
	Floods	0.996 ***	0.802 ***	0.935 ***	0.911 ***
	Earthquakes	-0.091	0.091	0.938 *	-0.071
	Storms	-0.093	-0.559 ***	0.838 *	-0.207

Note: The effects on growth are calculated using the coefficients reported in Table 2.
** significant at 10%; ** significant at 5%; *** significant at 1%*

In developing countries, a typical drought produces a reduction of agricultural and industrial annual growth rate of the order of 1 percentage point, leading to a decline of GDP growth by 0.6 percentage points per year or 3 percentage points over a period of 5 years.[20] This compares with an average annual per capita growth in developing countries of 1.35 percent during the 1961-2005 period. A typical flood increases growth in each major sector by about 0.8-0.9 percentage points, producing an increase of GDP growth by around 1 percentage point. A typical earthquake leads to a rise in industrial growth of about 0.9 percentage points, which, however, does not translate into an increase in aggregate growth. Finally, a typical storm has a dual effect, reducing agricultural growth by 0.6 percentage points and increasing industrial growth by 0.8 percentage points, which given the larger share of agriculture in developing economies results in a zero net effect on overall growth.

Clearly, the negative effects of droughts on aggregate and sectoral growth in developing countries can be substantial, while reports of moderate floods would in effect correspond to positive aggregate growth experiences (related to nationwide abundant rainfall). Moreover, given that the poverty to GDP elasticity is much larger for growth originating in agriculture (Christiaensen and Demery, 2007; Ligon and Sadoulet, 2007) and labor intensive (industrial) sectors (Loayza and Raddatz, 2006), the poor stand to be

[20] Note: $y_{\text{median drought}} - y_{\text{no drought}} = -0.076*(-5.90 - (-20)) = -1.071$

observed on agricultural growth (and also industry in developing countries). Storms also lower agricultural growth, but in developing countries, they also increase industrial growth. Similarly, earthquakes are found to bring about higher industrial growth in developing countries. In contrast, moderate (though not severe) floods have on average a positive effect on agricultural growth, and even other sectors of the economy, likely because localized flooding reflects broader nationwide abundance of rainfall.

Second, further underscoring the importance of disaggregation, while moderate disasters can have a positive growth effect on certain sectors, severe disasters don't. The impact of the 10 percent largest disasters in any category is found to be either insignificant or negative. When a natural disaster is severe enough, all the mechanisms that would potentially make it positive for growth are likely weakened. This also holds for severe floods and clarifies the seemingly surprising positive effect of floods. To the extent that reported floods are localized and reflective of more abundant national rainfall patterns, they would foster agriculture. Otherwise, the disruptions and damage caused by floods would cancel or outweigh the positive effects derived from plentiful rainfall.

Third, growth in developing countries is more sensitive to natural disasters—more sectors are affected and the magnitudes are non-trivial. This is consistent with the more marked presence of inter-sectoral linkages, following the more prominent role that agriculture plays in developing countries. Simulations indicate that a typical (median) drought reduces the annual per capita agricultural and industrial growth rate in developing countries by about 1 percentage point, together resulting in a reduction of annual per capital overall growth of 0.6 percentage points. A typical earthquake and storm increase industrial growth by about 1 percentage point each, consistent with the growth pattern predicted by theory when the capital labor ratio declines substantially, and further supported by the need for reconstruction following earthquakes and storms. As the elasticity of poverty to growth generated in agriculture and labor intensive sectors (such as construction and manufacturing) is substantially higher than the elasticity of poverty to growth generated outside agriculture in more capital intensive sectors, these results also suggest that the poor stand to be disproportionately affected.

Clearly, the time path of recovery and adjustment varies by shock and sector – and will likely be further affected by country-dependant institutional factors. Our findings